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(54) Title: A PROCESS OF MULTI LAYER PLASTIC RECYCLING

(57) Abstract: A process of recycling of multilayer plastic with the process of 1. Waste Segregation, Shredding Washing, Drying, blending or densifying or grinding and forming, granules Agglomeration. The waste is segregated and shredded in grinder machine, the wasted is washed in the washing plant and dried in high speed drier machine. The washed shredded material is agglomerated and heated and blending is done of molten multilayer plastic waste with specializes chemicals, the molten multilayer plastic waste is extruded to form recycled MLP granules.

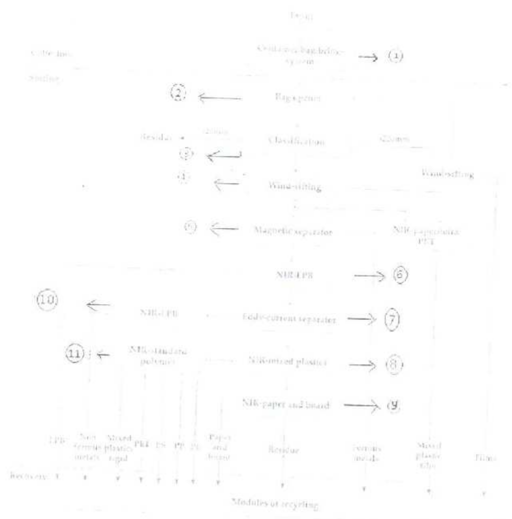


FIG.-1

Form 2

The Patent Act 1970

(39 of 1970)

COMPLETE SPECIFICATION

(Section 10; rule 13)

"A PROCESS OF MULTI LAYER PLASTIC RECYCLING"

We the partners of **M/S. THE SHAKTI PLASTIC INDUSTRIES** (A Partnership firm)having address at 15,GOVIND BHAVAN, S.V. ROAD, MALAD (WEST), MUMBAI-400 064

The following Specification particularly describes the nature of these invention and the Manner in which it is performed

TITLE :

A PROCESS OF MULTI LAYER PLASTIC RECYCLING"

FIELD OF THE IN INVENTION :-

The present invention relates to plastic recycling more particularly a process of recycling of multi layer plastic which is described with below mentioned embodiments.

BACKGROUND OF THE INVENTION:-

The invention relates to a process of recycling of multi layer plastic.

A) Multilayer Plastic Is A Combination Of any of LDPE (Low Density Polyethene), LLDPE (Linear Low Density Polythene), PP (Polypropene), HDPE (High density polythene), PET (Polyethylene tera phthalate, PS (poly styrene), PVC (polyvinyl chloride) PA (polyamide), poly foil and other plastics. Multilayer plastic materials are commonly used in order to combine the respective performance of different polymers. It is created to sufficiently protect sensitive products in order to obtain extended shelf life.

The main disadvantage of the multi layer plastic is its poor recyclability. According to the conventional method, there are two methods used for recycling multilayer plastic. The first way is to achieve the separation of different components either by delamination or the selective dissolution-repreipitation technique. The second method is to achieve recycling by combatibilization of non miscible polymer types.

The other disadvantage of recycling of multilayer plastic is high expenditure of energy and a limited scope.

The decent starting is a basic prerequisite for production of high quality recycled polymers. According to the prior art, multilayer plastic is not assigned to a special sorting stream. Instead multilayer plastic can be found in different recycling path ways like films, mixed plastics or the reside

The another object of the present invention is to make an eco friendly product for better environment protection.

OBJECT OF THE PRESENT INVENTION:

It is the object of the present invention is to develop a process of recycling of multi layer plastic. It is another object of the present invention is to develop a process of recycling of multi layer plastic with higher recyclability . It is another object of the present invention is to develop a process of multi layer plastic recycling with less expenditure of energy with a higher scope. It is another object of the present invention is to provide a better sorting system to recycle multi layer plastics.

BRIEF SUMMARY OF THE INVETION:

The present invention is a process of recycling of multi layer plastics by using specialized chemicals.

TECHNICAL DETAILS OF THE INVENTION:

According to the present inventions the multi player plastic waste or scrap which are suitable for recycling or reprocessing are selected.

The multilayer plastics waste is segregated as per the codes 1-7 mentioned in the BIS guide lines (IS-14534:1998).

There after the multiplayer plastic waste us shredded. Shredding systems are used to reduce the size of a given material. A reliable and sturdily industrial shredder is used for shredding of the multi layer plastic waste.

The shredded plastic waste is routed washing tanks filled with cold water containing no additives. After immersion, the materials are removed to a horizontal centrifuge in order to remove impurities. Then the same is send to under water grinding phase. After this the material goes to vertical centrifuge where the friction from knives removes any final residues.

The multi layer plastic waste is dried in high speed drier machine. There after the blending is done of molten multi layer plastic waste with specialised chemicals. The molten multi layer plastic waste is extruded to form recycled multi layer plastic granules.

The other method of multi layer plastic recycling is after washing the multi layer plastic waste is densified. There after the multilayer plastic waste is grinded and granules are produced.

The third method of multi layer plastic waste is shredded. During the production and process waste occurs which can be agglomerated continuously with the plast- agglomerate in to free-flowing granules and reused for production as a valuable raw material. The agglomerated material is then heated in extruded where blending is done of molten multi layer plastic waste with specialized chemicals. The molter multi layer plastic waste is extruded to form recycled multi layer plastic granules.

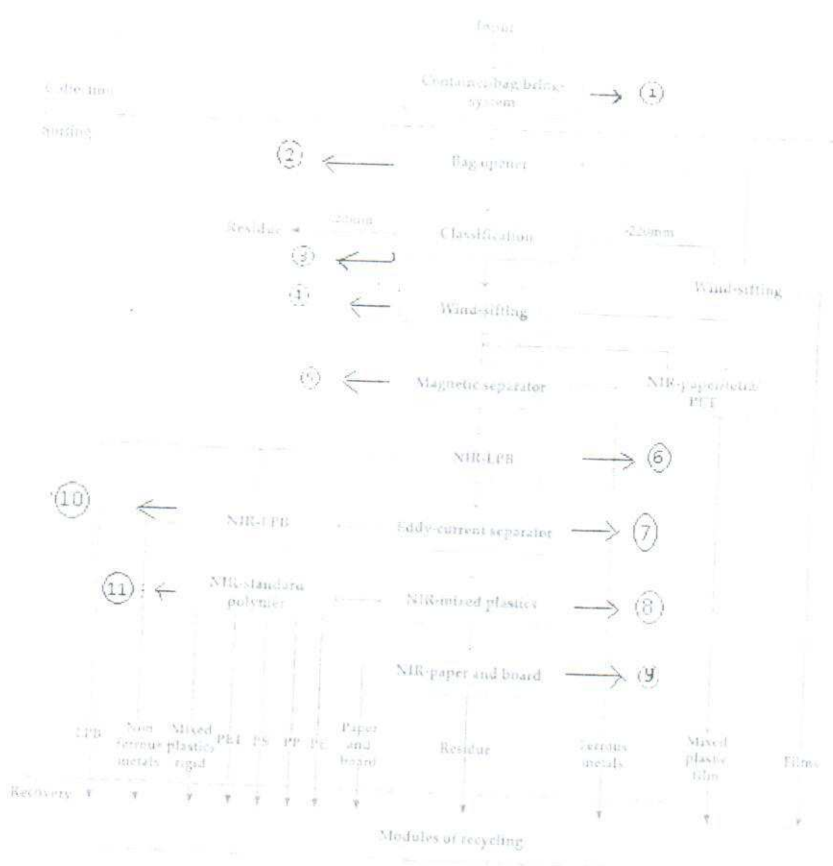
Even though there has been describe above multi layer plastic recycling process in accordance with present invention, it is to be understood that the invention, it is not limited there to consequently, any and all variations and equivalent arrangements which my occur to those skilled in the applicable area to be considered to be within the scope and spirit of the invention as set forth in the calms, which are appended here to as part of this application.

BRIEF DESCRIPTION OF THE DRAWINGS:

Figure 1 is the modules of recycling prior art.

Figure-2 is the schematic overview of the recycling methods of prior Art.

Figure-3 is the process chart of multi layer plastic recycling as per the present invention.



Input
Sorting
Bag opener
Residue
Classification
Wind-sifting
Wind-sifting
Magnetic separator
NIR-LPB
NIR-LPB
Eddy-current separator
NIR-standard
polymer
NIR-mixed plastics
NIR-paper and board
Non
ferrous
metals
Mixed
plastics
rigid
PET
PS
PP
PE
Paper
and
board
Residue
Mixed
plastic
film
Recovery
Modules of recycling

FIG.-1

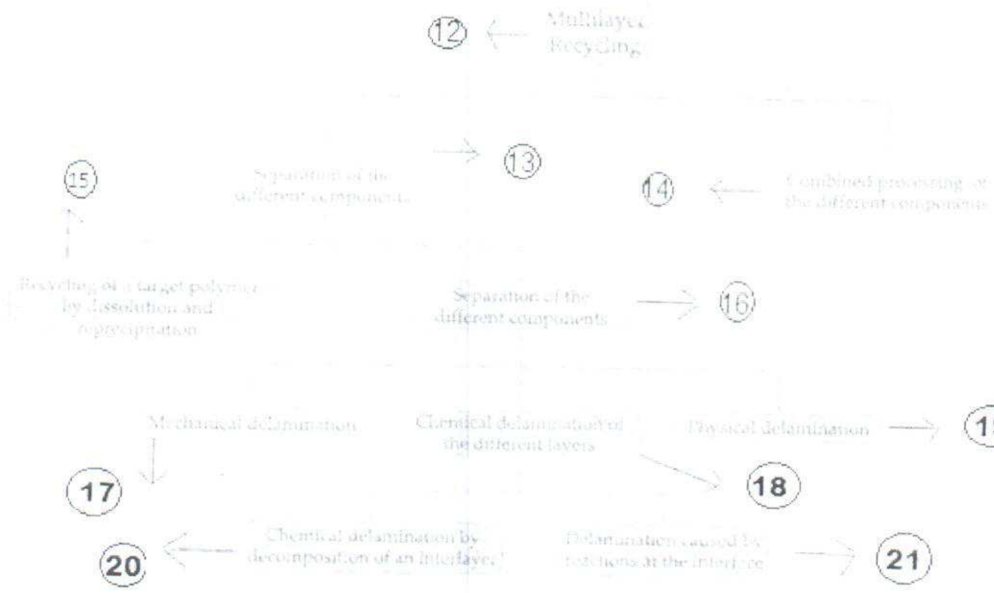

12
Multilayer Recycling
13
Separation of the different components
14
Combined processing of the different components
15
Recycling of a target polymer by dissolution and reprecipitation
Separation of the different components
16
Mechanical delamination
Chemical delamination of the different layers
Physical delamination
19
17
18
20
Chemical delamination by decomposition of an Interlayer
21

FIG-2

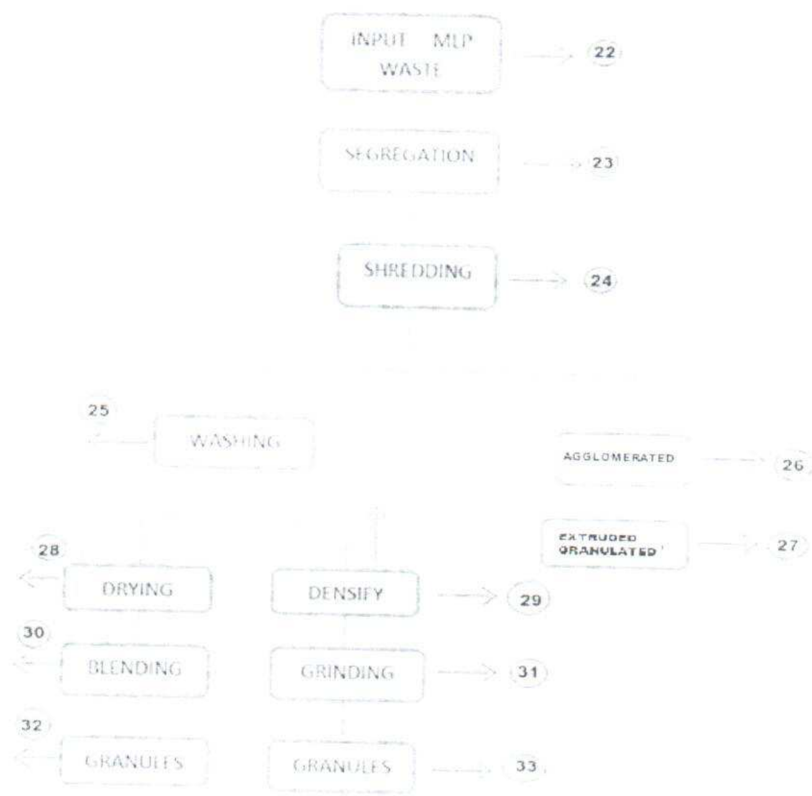


FIG-3

DETAIL DESCRIPTION OF DRAWINGS:

Referring to the drawing figure 1 depict the Container/bag/bring system 1, Bag opener 2, Classification 3, Wind-sifting 4, magnetic separator 5, NIR-LPB 6, Eddy-current separator 7, NIR-mixed plastics 8, NIR-paper and board 9, NIR-LPB 10, NIR-paper and board 11

Referring to the drawing figure 2 depict multilayer recycling 12, separation of the different components 13, combined processing of the different components14, Recycling of a target polymer by dissolution and reprecipitation 15, separation of the different components 16, mechanical delamination 17, chemical delamination of the different layer 18, physical delamination 19, chemical delamination by decomposition of an interlayer 20, delamination caused by reactions at the interface 21.

Referring to the drawing figure 3 INPUT MLP WASTE 22, SEGREGATION 23, SHREDDING 24, WASHING 25, AGGLOW 26, GRANULES 27, DRYING 28, DENSIFY 29, BLENDING 30, GRINDING 31, GRANULES 32, GRANULES 33.

I CLAIM

A process of recycling of multilayer plastic with the process of :-

A) Selection and Segregation of plastic waste
B) Shredding of plastic waste
C) Washing of plastic waste
D) Drying of plastic waste
E) Blending of plastic waste
F) Densify
G) Grinding
H) Agglow meration
I) Extrusion
J) Granules

1) A Process as claimed in claim – 1 wherein the multi layer plastic waste is segregated and shredded in grinder machine.

2) A process as claimed in claim 1 or 2 wherein the waste is washed in the washing plant filled with cold water and dried in high speed drier machine.

3) A process as claimed in claim 1 to 3 whererin washed shredded material is blended with specialised chemicals.

4) A process as claimed in claim 1 to 4 wherein the molton multilayer plastic waste is extruded to form recycled multi layer plastic granules.

5) A process as claimed in claim 1 to 5 wherein the process can alternatively performed after washing the multilayer plastic waste, the same is grinded and granules are produced.

6) A process as claimed in claim 1 to 5 can be performed alternatively by shredding the plastic waste and the waste is agglomerated continuously with the agglomerator into free-flowing granules. The agglomerated material is heated, blended and extruded to form multilayer plastic granules.

7) A process of recycling as claimed here in above in claim 1 to 7 is hereby illustrated with the help of accompanying drawings.

Dated 4th DAY OF JANUARY, 2019

Signature Joseph

ADV.JOSEPH VARIKASERY

FOR, VARIKASERY AND VARIKASERY

(Agent for the Applicant)

ABSTRACT :

A process of recycling of multilayer plastic with the process of 1.Waste Segregation, Shredding Washing, Drying, blending or densifying or grinding and forming, granules Agglomeration. The waste is segregated and shredded in grinder machine, the wasted is washed in the washing plant and dried in high speed drier machine, The washed shredded material is agglomerated and heated and blending is done of molten multilayer plastic waste with specializes chemicals, the molten multilayer plastic waste is extruded to form recycled MLP granules.